

RAMESWAR BHATTACHARJEE

Electronic Structure Theory • π -Conjugated and Radical Materials • Machine Learning for Chemical Discovery

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RESEARCH PROFILE

Computational chemist developing predictive electronic-structure and machine-learning frameworks for electronically complex molecular and materials systems. My research connects orbital symmetry, topology, and spin/charge delocalization to structure–property relationships in π -conjugated polymers, open-shell radical assemblies, and catalytic reaction networks. Recent independent work establishes design principles for topological conjugated polymers, quantifies non-classical “pancake” bonding in radical π -stacks, and introduces interpretable descriptor-based machine learning that predicts DFT-level electronic properties across thousands of radical dimer packings.

I am seeking a tenure-track faculty position in which to build an independent program on predictive design of open-shell and topological organic electronic materials, combining high-level electronic structure theory with data-driven discovery.

RESEARCH METRICS

37	1,170+	19	17	7	3
Peer-reviewed publications	Citations	h-index	First / co-first author	Corresponding author	Journal cover features

Citation metrics from Google Scholar, July 2026. Six papers as both first and corresponding author; one additional paper as designated co-first author (equal contribution).

ACADEMIC APPOINTMENTS

Research Assistant Professor	June 2025 – Present
Georgetown University, Department of Chemistry — independent research on topological conjugated polymers, radical π -stacking, and machine-learning-guided materials prediction.	
Postdoctoral Fellow	Nov. 2022 – May 2025
Georgetown University — with Prof. Miklos Kertesz. Electronic structure of π -conjugated polymers and open-shell radical aggregates.	
Postdoctoral Fellow	Dec. 2020 – Oct. 2022
University of South Dakota — with Prof. Pere Miró. Actinide speciation and polyoxocationic cluster chemistry.	
Postdoctoral Fellow	Jan. 2019 – Dec. 2020
University of Delaware — with Prof. Dion Vlachos and Dr. Stavros Caratzoulas. Computational catalysis for biomass valorization (DOE Energy Frontier Research Center).	

EDUCATION

Ph.D., Chemistry	2018
Indian Association for the Cultivation of Science (IACS), Kolkata. Advisor: Prof. Ayan Datta. Thesis: Computational Studies of Homogeneous and Heterogeneous Catalysis.	
M.Sc., Chemistry	2012
Indian Institute of Technology (IIT) Guwahati.	
B.Sc., Chemistry	2010
University of Burdwan.	

FEDERAL RESEARCH FUNDING — CONTRIBUTING PERSONNEL

Named project participant on active federal awards, as reported to the funding agencies by the Principal Investigators.

NSF CHE-2505194	Sept. 2025 – Aug. 2028
Collaborative Research: Radical and Biradical Polycyclic Aromatic Hydrocarbons in Complex Aggregation Environments.	

Division of Chemistry, Macromolecular/Supramolecular/Nanochemistry Program. PI: M. Kertesz (Georgetown); collaborative with H. Lischka (Texas Tech). Total award: \$323,702.

- Role: NSF-reported project participant. **First author on two of the four publications formally reported as scientific outputs of this award** (*ACS Mater. Au* 2025; *ACS Phys. Chem. Au* 2026).

DOE DE-SC0019017

2023 – Present

Exploration of Radial Conjugation Pathways in π -Electron Materials.

U.S. Department of Energy, Office of Science. PI: J. D. Tovar (Johns Hopkins University).

- Role: DOE-reported project participant. Corresponding author on resulting publication, *Chem. Sci.* 2025, 16, 14595.

Independent research proposals authored

- Ramanujan Fellowship, Government of India (submitted 2026) — independent faculty research program on computationally guided design of conjugated materials for optoelectronic applications.
- Alexander von Humboldt Fellowship (2022) — theoretical strategies for selective heavy-metal separation from groundwater systems.

SELECTED RESEARCH CONTRIBUTIONS

Topological electronic structure in π -conjugated materials

- *J. Am. Chem. Soc.* 2024 — Strain-induced topological transition in π -conjugated polymers
- *Chem. Sci.* 2025 — Quinonoid radial π -conjugation
- *Chem. Mater.* 2024 — Continuous bandgap tuning through mechanical strain

Established electronic topology, quinonoid–aromatic competition, and mechanical strain as tunable design parameters for controlling band structure and emergent electronic phases in polymeric semiconductors.

Radical π -stacking and non-classical pancake bonding

- *J. Am. Chem. Soc.* 2024 — Charge effects in π -stacked perylenes
- *Chem. Sci.* 2024 — Trimeric triphenylene radical cation aggregation and bonding
- *ACS Mater. Au* 2025 — Pancake bonding in cationic acene dimers

Quantified how charge distribution, orbital symmetry, and spin/charge delocalization govern intermolecular stabilization in stacked radical assemblies, in sustained collaboration with crystallographic groups that validated the predictions experimentally.

Machine learning for structure–property prediction in open-shell materials

- **Descriptor-based ML across 2,582 π -radical dimer geometries** spanning five PAH radical systems, linking molecular packing to DFT-derived electronic structure.
- Nineteen chemically interpretable geometric descriptors predict fragment-orbital SOMO coupling, frontier-orbital gap, and interaction energies with test-set $R^2 = 0.94$ – 0.98 (0.82 – 0.92 under held-out geometry clusters).
- Validated against system-mean baselines and 500-fold Y-randomization controls, separating genuine geometric information from family-level correlation — a rigor standard often missing in chemical ML.
- Datasets and analysis code released openly (Zenodo DOI: 10.5281/zenodo.21712041).

Demonstrates that interpretable, physically grounded descriptors can replace expensive broken-symmetry DFT for rapid screening of radical packing motifs — the methodological foundation of my proposed independent program.

Computational catalysis and reaction mechanism discovery

- *Green Chem.* 2020 — Biomass-derived lubricant base oil synthesis
- *Chem. Sci.* 2019; *Angew. Chem. Int. Ed.* 2016 — Metal-free CO₂ reduction chemistry
- *Chem. Eur. J.* 2017 — Direct and autocatalytic reductive elimination

Advanced mechanistic understanding of homogeneous and heterogeneous reaction landscapes relevant to sustainable synthesis and CO₂ transformation.

COMPUTATIONAL EXPERTISE

- **Molecular electronic structure:** Gaussian, ORCA, Q-Chem. Ground- and excited-state DFT, TDDFT, broken-symmetry DFT for open-shell and antiferromagnetically coupled systems, fragment-orbital DFT for electronic coupling and transfer integrals.
- **Periodic and solid-state:** VASP, CP2K, Quantum ESPRESSO. Band structure, density of states, strain-dependent electronic structure of one-dimensional polymers, surface adsorption.

- **Multireference methods:** COLUMBUS, MOLCAS/OpenMolcas, Molpro. Multireference treatments of biradical and polyradical character in open-shell PAHs, developed through collaboration with the Lischka group.
- **Conformational sampling and dynamics:** GOAT conformer search, GFN2-xTB semiempirical tight-binding, automated high-throughput generation and curation of multi-thousand-structure datasets.
- **Machine learning and data science:** Python, scikit-learn, PyTorch. Ensemble tree methods (extremely randomized trees), multilayer perceptron regression, descriptor engineering, cross-validation and cluster-based validation design, permutation and impurity feature-importance analysis, Y-randomization controls.
- **Analysis and bonding theory:** Natural bond orbital analysis, energy decomposition, Mulliken/Löwdin population analysis, spin-density and pancake-bonding characterization.
- **High-performance computing:** Georgetown University High Performance Computing cluster. Routine management of large-scale parallel quantum-chemical workflows, including automated generation and analysis of multi-thousand-structure datasets.
- **Programming and workflow:** Python, Bash, Git; version-controlled and openly archived analysis pipelines.

PEER-REVIEWED PUBLICATIONS

37 publications, reverse chronological order. Applicant name in bold. Asterisk (*) indicates corresponding author; double dagger (‡) indicates equal contribution.

1. Reid, H.W.; **Bhattacharjee, R.***; Sun, P.B.; Skala, M. E.; Serck, N. P.; Golder, M. R.*; Kertesz, M.; Jasti, R.*. High Molecular Weight [n]Cycloparaphenylene Conjugated Polymers and Copolymers with Improved Electron Delocalization. *Macromolecules* 2026, 59, 6963–6972. DOI: 10.1021/acs.macromol.6c00539
2. **Bhattacharjee, R.***; McCormack, M.;‡ Zhu, Y.; Wei, Z.; Kertesz, M.*; Petrukhina, M.A.* Chemical Oxidation of Coronene: A Crystallographic and Computational Study of Charge-Regulated π -Stacking. *ACS Phys. Chem. Au* 2026, 571-579. DOI: 10.1021/acspyschemau.6c00020
3. Yin, G.* **Bhattacharjee, R.**; Wang, T.; Kertesz, M. Wilson polygons and the topology of zero-dimensional systems. *Phys. Rev. B* 2026, 113, 115104. DOI: 10.1103/7wx3-6syf
4. McCormack, M.; **Bhattacharjee, R.**; Wei, Z.; Petrukhina, M.A.* Chemical Oxidation of Chrysene: A Structural and Theoretical Description of a Mixed-Valent Trimeric Radical Cation. *Cryst. Growth Des.* 2025, 25, 9549–9554. DOI: 10.1021/acs.cgd.5c01176
5. McCormack, M.;‡ **Bhattacharjee, R.**;‡ Wei, Z.; Kertesz, M.*; Petrukhina, M.A.* Building a Mixed Polyarene π -Stack through Chemical Oxidation: A Structural and Theoretical Investigation. *Precis. Chem.* 2025, 3, 631–639. DOI: 10.1021/prechem.5c00066
6. **Bhattacharjee, R.***; Tovar, J.D; Kertesz, M.* Quinonoid radial π -conjugation. *Chem. Sci.* 2025, 16, 14595. DOI: 10.1039/D5SC00639B
7. **Bhattacharjee, R.***; Lischka, H.; Kertesz, M.* Pancake Bonding in the Stabilization of Cationic Acene Dimers *ACS Mater. Au* 2025, 5, 365–376. DOI: 10.1021/acsmaterialsau.4c00153
8. **Bhattacharjee, R.***; Kertesz, M.* Topological transition in aromatic and quinonoid π -conjugated polymers induced by static strain. *J. Am. Chem. Soc.* 2024, 146, 26497–26504. DOI: 10.1021/jacs.4c10064
9. **Bhattacharjee, R.**;‡ McCormack, M.;‡ Jervis, H.; Wei, Z.; Petrukhina, M.A.*; Kertesz, M.* A unique trimeric triphenylene radical cation: Stacking aggregation, bonding, and stability. *Chem. Sci.* 2024, 15, 15221-15231. DOI: 10.1039/d4sc03774j
10. **Bhattacharjee, R.***; Jervis, H.; McCormack, M.; Petrukhina, M.A.; Kertesz, M.* Structure and Bonding in π -Stacked Perylenes: The Impact of Charge on Pancake Bonding. *J. Am. Chem. Soc.* 2024, 146, 10465–10477. DOI: 10.1021/jacs.3c14065
11. **Bhattacharjee, R.***; Kertesz, M.* Continuous Topological Transition and Bandgap Tuning in Ethynyl-Linked Acene π -Conjugated Polymers through Mechanical Strain. *Chem. Mater.* 2024, 36, 1395–1404. DOI: 10.1021/acs.chemmater.3c02547
12. Ren, L.; Xiao, Y.; **Bhattacharjee, R.**; Wu, J.; Tang, P.; Caratzoulas, S.; Meng, C.; Guo, Q.*; Tsapatsis, M.* High Yield of L-Sorbose via D-Glucose Isomerization in Ethanol over a Bifunctional Titanium-Boron-Beta Zeolite. *Chem. Eur. J.* 2024, 30, e202402341 DOI: 10.1002/chem.202402341
13. McCormack, M.; **Bhattacharjee, R.**; Jervis, H.; Wei, Z.; Kertesz, M.*; Petrukhina, M.A.* Structural and Computational Insights into Partially Oxidized π -stacked Perylene. *Cryst. Growth Des.* 2023, 23 7496-7503. DOI: 10.1021/acs.cgd.3c00912
14. **Bhattacharjee, R.**; Hare, E; Miró, P.* Structure, Properties, and Reactivity of Polyoxocationic Zirconium and Hafnium Clusters: A Computational Investigation *Inorg. Chem.* 2023, 62, 5081-5087. DOI: 10.1021/acs.inorgchem.2c04142
15. **Bhattacharjee, R.**; Miró, P.* Aqueous Speciation of Tetravalent Actinides in the Presence of Chloride and Nitrate Ligands *Inorg. Chem.* 2022, 37, 14718-14725. DOI: 10.1021/acs.inorgchem.2c02064

16. Deegan, M. M.; **Bhattacharjee, R.**; Caratzoulas, S.; Bloch, E. D.* Stabilizing Porosity in Organic Cages Through Coordination Chemistry. *Inorg. Chem.* 2021, 60, 7044– 7050. DOI: 10.1021/acs.inorgchem.0c03590
17. Liu, S; **Bhattacharjee, R.**; Li S; Danielson, A; Mazal, T; Saha, B.*; Vlachos, D.G.* Thiol-promoted catalytic synthesis of high-performance furan-containing lubricant base oils from biomass-derived 2-alkylfurans and ketones. *Green Chem.* 2020, 22, 7896-7906. DOI: 10.1039/D0GC01897J
18. Rowland, C. A; Lorzing, G. R; **Bhattacharjee, R.**; Caratzoulas, S; Yap, G; Bloch, E. D.* Design and synthesis of aryl-functionalized carbazole-based porous coordination cages. *Chem. Commun.* 2020, 56, 9352-9355. DOI: 10.1039/D0CC03910A
19. Anju, B. S.; **Bhattacharjee, R.**; Ahammad, S.; Datta, A.*; Kundu, S.* Deoxygenation of Nitrosoarene by N- Heterocyclic Carbene (NHC): Elusive Breslow-type Intermediate Bridging Carbene and Nitrene. *Chem. Commun.* 2020, 56, 12166-12169. DOI: 10.1039/D0CC05192F
20. Lorzing, R. G.; Gosselin, E. J.; Lindner, B. S.; **Bhattacharjee, R.**; Yap, G.; Caratzoulas, S.; Bloch, E. D.* Design and synthesis of capped-paddlewheel-based porous coordination cages. *Chem. Commun.* 2019, 55, 9527-9530. DOI: 10.1039/C9CC05002G
21. **Bhattacharjee, R.**; Majumdar, T.; Datta, A.* Analysis of pseudo jahn–teller distortion based on natural bond orbital theory: Case study for silicene. *J. Comput. Chem.* 2019, 40, 1488-1495. DOI: 10.1002/jcc.25815
22. Sau, S. C.; **Bhattacharjee, R.**; Vardhanapu, P. K.; Hota, P.K.; Vijaykumar, G.; Govindarajan, R.; Datta, A.*; Mandal, S. K.* Transforming atmospheric CO₂ into alternative fuels: a metal-free approach under ambient conditions. *Chem. Sci.* 2019, 10, 1879–1884. DOI: 10.1039/C8SC03581D
23. Rath, S. P.; Sengupta, D.; Ghosh, P.; **Bhattacharjee, R.**; Chakraborty, M.; Samanta, S.; Datta, A.; Goswami, S.* Effects of Ancillary Ligands on Redox and Chemical Properties of Ruthenium Coordinated Azoaromatic Pincer. *Inorg. Chem.* 2018, 57, 11995–12009. DOI: 10.1021/acs.inorgchem.8b01558
24. **Bhattacharjee, R.**; Datta, A.* Understanding Thermal and Photochemical Aryl-Aryl Cross-coupling using metal complex. *Chem. Eur. J.* 2018, 24, 13636-13646. DOI: 10.1002/chem.201802634
25. Pramanick, R.; **Bhattacharjee, R.**; Sengupta, D.; Datta, A.*; Goswami, S.* An azoaromatic ligand as four electron four proton reservoir: catalytic dehydrogenation of alcohols by its Zinc(II) complex. *Inorg. Chem.* 2018, 57, 6816-6824. DOI: 10.1021/acs.inorgchem.8b00034
26. **Bhattacharjee, R.**; Datta, A.* Role of Carbon support for subnanometer gold-cluster catalyzed disiloxane synthesis from hydrosilane and water. *J. Phys. Chem. C* 2017, 121, 20101-20112. DOI: 10.1021/acs.jpcc.7b04419
27. **Bhattacharjee, R.**; Nijamudheen, A.; Datta, A.* Direct and autocatalytic reductive elimination from gold complexes ([(Ph₃P)Au(Ar)(CF₃)(X)], X= F, Cl, Br, I): The key role of halide ligands. *Chem. Eur. J.* 2017, 23, 4169-4179. DOI: 10.1002/chem.201605784
28. **Bhattacharjee, R.**; Datta, A.* Supported sub-nanometer gold cluster catalyzed transfer hydrogenation of aldehydes to alcohols. *J. Phys. Chem. C* 2016, 120, 24449-24456. DOI: 10.1021/acs.jpcc.6b07520
29. **Bhattacharjee, R.**; Nijamudheen, A.; Karmakar, S.; Datta, A.* Strain control: Reversible H₂ activation and H₂/D₂ exchange in Pt complexes. *Inorg. Chem.* 2016, 55, 3023-3029. DOI: 10.1021/acs.inorgchem.5b02904
30. Sau, S. C.; **Bhattacharjee, R.**; Vardhanapu, P. K.; Vijaykumar, G.; Datta, A.*; Mandal, S. K.* Metal-free reduction of CO₂ to methoxyborane under ambient conditions through borondiformate formation. *Angew. Chem. Int. Ed.* 2016, 128, 15371-15375. DOI: 10.1002/anie.201609040
31. Asha, K.; **Bhattacharjee, R.**; Mandal, S.* Complete transmetalation in a metal–organic framework by metal ion metathesis in a single crystal for selective sensing of phosphate ions in aqueous media. *Angew. Chem. Int. Ed.* 2016, 128, 11700-11704. DOI: 10.1002/anie.201606185
32. Sengupta, D.; **Bhattacharjee, R.**; Pramanick, R.; Rath, S. P.; Saha Chowdhury, N.; Datta, A.*; Goswami, S.* Exclusively ligand-mediated catalytic dehydrogenation of alcohols. *Inorg. Chem.* 2016, 55, 9602-9610. DOI: 10.1021/acs.inorgchem.6b01310
33. Roy, S.; Baral, A.; **Bhattacharjee, R.**; Jana, B.; Datta, A.; Ghosh, S.; Banerjee, A.* Preparation of multi-coloured different sized fluorescent gold clusters from blue to NIR, structural analysis of the blue emitting Au₇ cluster, and cell- imaging by the NIR gold cluster. *Nanoscale* 2015, 7, 1912-1920. DOI: 10.1039/C4NR04338C
34. Pratik, S. M.; Chowdhury, C.; **Bhattacharjee, R.**; Jahiruddin, S.; Datta, A.* Pseudo Jahn–Teller distortion for a tricyclic carbon sulfide (C₆S₈) and its suppression in S-oxygenated dithiine (C₄H₄(SO₂)₂). *Chem. Phys.* 2015, 460, 101-105. DOI: 10.1016/j.chemphys.2015.04.009
35. **Bhattacharjee, R.**; Nijamudheen, A.; Datta, A. Mechanistic insights into the synergistic catalysis by Au (I), Ga (III), and counterions in the Nakamura reaction. *Org. Biomol. Chem.* 2015, 13, 7412-7420. DOI: 10.1039/C5OB00626K
36. Nijamudheen, A.; **Bhattacharjee, R.**; Choudhury, S.; Datta, A.* Electronic and chemical properties of germanene: The crucial role of buckling. *J. Phys. Chem. C* 2015, 119, 3802–3809. DOI: 10.1021/jp511488m

37. Maiti, D. K.; **Bhattacharjee, R.**; Datta, A.*; Banerjee, A.* Modulation of fluorescence resonance energy transfer efficiency for white light emission from a series of stilbene-perylene based donor-acceptor pair. *J. Phys. Chem. C* 2013, 117, 23178-23189. DOI: 10.1021/jp409042p

MANUSCRIPTS UNDER REVIEW AND IN PREPARATION

1. **Bhattacharjee, R.***; Lischka, H.; Kertesz, M.* How Molecular Packing Controls Electronic Structure in π -Stacked Radical Dimers: A DFT and Descriptor-Based Machine-Learning Study. *Submitted to Journal of Chemical Information and Modeling*, 2026. Preprint posted. Data and analysis code openly archived: Zenodo DOI 10.5281/zenodo.21712041.
2. **Bhattacharjee, R.***; Lischka, H.; Kertesz, M.* Mapping the π - σ Bonding Landscape in a Pancake-Bonded Radical Dimer via Data-Driven and Chemistry-Guided Analysis. *Manuscript in final preparation*, 2026.

CONFERENCE PRESENTATIONS

- Machine learning of packing-electronic structure relationships in π -stacked radical dimers. *Gordon Research Conference on Computational Materials Science and Engineering*, Sunday River, Newry, ME, Aug. 2-7, 2026. Poster.
- Topological conjugated polymers under mechanical strain and a prospect on machine learning. *Third International Symposium on Machine Learning in Quantum Chemistry*, Knoxville, TN, Oct. 5-8, 2025. Poster.
- Topological phases of conjugated polymers under mechanical strain. *ACS National Meeting, Fall 2025*, Washington, DC, Aug. 20, 2025. Oral.
- Mechanical and chemical tuning of the band gap of linearly bridged one-dimensional acene polymers. *ACS National Meeting, Fall 2023*, San Francisco, CA, Aug. 17, 2023. Oral.
- Speciation of AnCl_6^{2-} (An = U, Np, Pu) in aqueous solution. *ACS National Meeting, Spring 2022*, San Diego, CA, Mar. 20, 2022. Oral.

TEACHING AND MENTORING

Teaching experience

Faculty Assistant — Physical Chemistry I

Georgetown University.

Fall 2025

Teaching Assistant — Computational Chemistry Methods

Georgetown University.

Spring 2025

Teaching Assistant — Physical Chemistry I

Georgetown University.

Fall 2024

Courses prepared to teach

- **Undergraduate:** Physical Chemistry I and II (thermodynamics, kinetics, quantum chemistry), General Chemistry.
- **Graduate:** Computational Chemistry Methods, Electronic Structure Theory, Chemical Bonding and Group Theory.
- **Proposed new course:** Machine Learning for Molecular and Materials Discovery — a project-based course pairing electronic structure fundamentals with descriptor design, model validation, and reproducible open-science practice.

Research mentoring

- Four undergraduate researchers mentored since 2019, across two institutions.
- Georgetown University (2022-present): mentored an undergraduate researcher to **two co-authored publications** in *J. Am. Chem. Soc.* and *Cryst. Growth Des.*; currently mentoring a second undergraduate researcher.
- University of Delaware (2019-2020): mentored two undergraduate researchers through the NSF Research Experiences for Undergraduates (REU) program.
- Training focus: electronic structure theory, quantum-chemical workflow design, high-performance computing practice, and reproducible computational analysis.

AWARDS AND HONORS

- EFRC Postdoctoral Fellowship, Catalysis Center for Energy Innovation (DOE Energy Frontier Research Center), University of Delaware, 2019-2020.
- Senior Research Fellowship, Council of Scientific and Industrial Research (CSIR), India, 2015-2018.
- Junior Research Fellowship, Council of Scientific and Industrial Research (CSIR), India, 2013-2015.
- CSIR National Eligibility Test (NET), Chemical Sciences — All-India Rank 44, 2012.
- Graduate Aptitude Test in Engineering (GATE), Chemical Sciences — All-India Rank 167, 2012.

- Three journal cover features.

PROFESSIONAL SERVICE

- Peer reviewer, **50+ manuscript reviews completed**, primarily for American Chemical Society and MDPI journals; additional reviews for Springer Nature and Wiley titles.
- Reviewer Board Member, *Inorganics* (MDPI).
- Postdoctoral Symposium Judge, Georgetown University.
- Member, American Chemical Society.
- Open science: public release of computational datasets and machine-learning analysis code via Zenodo.

REFERENCES

Prof. Miklos Kertesz — *Postdoctoral advisor and current supervisor*

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Prof. Hans Lischka — *Collaborator, NSF CHE-2505194*

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